

Vector Integration: Evaluating a Line Integral Along a Curved Path

1. Line Integral Parameterization Using an Existing Coordinate

Name of Concept

Line Integration along an Intersecting Space Curve parameterized by a single coordinate variable (x).

Core Mathematical Formulas

The expansion of the vector line integral into its scalar coordinate components is:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (f_1 dx + f_2 dy + f_3 dz)$$

The given constraint equations of the space curve path C are:

$$y = \frac{x^2}{4}, \quad z = \frac{3x^3}{8}$$

Beginner-Friendly Explanation of Operations

- Choosing x as the Parameter:** In the previous problem, we introduced a completely separate variable t . However, if the path constraints already express y and z directly as functions of x , we can bypass t entirely. We choose x itself as our primary parameter.
- Expressing the Dependent Variables:** Because y and z are bound to the behavior of x along this curved track, we rearrange the given path equations to isolate y and z :
 - y becomes $\frac{x^2}{4}$
 - z becomes $\frac{3x^3}{8}$
- Converting the Differentials (dy and dz):** We differentiate our expressions for y and z with respect to x using standard power rules to establish equations for the differential steps dy and dz in terms of dx :
 - Differentiating $y = \frac{1}{4}x^2$ yields $\frac{dy}{dx} = \frac{2x}{4} = \frac{x}{2}$, meaning $dy = \frac{x}{2} dx$.
 - Differentiating $z = \frac{3}{8}x^3$ yields $\frac{dz}{dx} = \frac{9x^2}{8}$, meaning $dz = \frac{9x^2}{8} dx$.
- Simplification to a Single-Variable Integral:** By substituting these equations into our line integral layout, every instance of y , z , dy , and dz is replaced with an equivalent expression containing only x and dx . This reduces the problem to a standard definite single-variable calculus integral from the starting limit $x = 0$ to the ending limit $x = 2$.

2. Step-by-Step Problem Solution

Question

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the curve defined by $x^2 = 4y$ and $3x^3 = 8z$ from $x = 0$ to $x = 2$.

Step 1: Identify components and map path constraints

Extract the components of the vector field $\vec{F}$:

$$f_1 = 3x^2$$

$$f_2 = 2xz - y$$

$$f_3 = z$$

Isolate y and z from the given curve boundary equations:

$$y = \frac{x^2}{4}$$
$$z = \frac{3x^3}{8}$$

Step 2: Calculate the coordinate differentials in terms of dx

Take the derivative of each isolated coordinate equation with respect to x :

$$\frac{dy}{dx} = \frac{2x}{4} = \frac{x}{2} \implies dy = \frac{x}{2} dx$$
$$\frac{dz}{dx} = \frac{3(3x^2)}{8} = \frac{9x^2}{8} \implies dz = \frac{9x^2}{8} dx$$

Step 3: Rewrite vector components purely in terms of x

Substitute the expressions for y and z into the components f_2 and f_3 :

$$f_1 = 3x^2$$
$$f_2 = 2x \left(\frac{3x^3}{8} \right) - \left(\frac{x^2}{4} \right) = \frac{6x^4}{8} - \frac{x^2}{4} = \frac{3x^4}{4} - \frac{x^2}{4}$$
$$f_3 = \frac{3x^3}{8}$$

Step 4: Substitute all expressions into the line integral formula

Assemble the total scalar integral component structure:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 [(f_1) dx + (f_2) dy + (f_3) dz]$$

Substitute the component functions and their corresponding differentials:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \left[(3x^2) dx + \left(\frac{3x^4}{4} - \frac{x^2}{4} \right) \left(\frac{x}{2} dx \right) + \left(\frac{3x^3}{8} \right) \left(\frac{9x^2}{8} dx \right) \right]$$

Step 5: Expand and distribute the terms inside the brackets

Multiply out the terms within each individual component group:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \left[3x^2 dx + \left(\frac{3x^5}{8} - \frac{x^3}{8} \right) dx + \left(\frac{27x^5}{64} \right) dx \right]$$

Step 6: Combine like algebraic terms over a common denominator

Group all terms under a single factor of dx :

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \left[3x^2 - \frac{x^3}{8} + \left(\frac{3x^5}{8} + \frac{27x^5}{64} \right) \right] dx$$

Find a common denominator of 64 for the x^5 terms to merge them:

$$\frac{3x^5}{8} = \frac{3 \cdot 8 \cdot x^5}{8 \cdot 8} = \frac{24x^5}{64}$$
$$\frac{24x^5}{64} + \frac{27x^5}{64} = \frac{51x^5}{64}$$

Substitute this simplified term back into the complete integrand:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^2 \left[3x^2 - \frac{x^3}{8} + \frac{51x^5}{64} \right] dx$$

Step 7: Compute the definite single-variable integral

Apply standard calculus integration rules ($\int x^n dx = \frac{x^{n+1}}{n+1}$) component by component:

$$\int_C \vec{F} \cdot d\vec{r} = \left[3 \left(\frac{x^3}{3} \right) - \frac{1}{8} \left(\frac{x^4}{4} \right) + \frac{51}{64} \left(\frac{x^6}{66} \right) \right]_0^2$$

Simplify the constant fraction coefficients before substituting values:

$$\int_C \vec{F} \cdot d\vec{r} = \left[x^3 - \frac{x^4}{32} + \frac{51x^6}{384} \right]_0^2$$

Reduce the fraction $\frac{51}{384}$ by dividing the numerator and denominator by 3:

$$\frac{51 \div 3}{384 \div 3} = \frac{17}{128}$$

$$\int_C \vec{F} \cdot d\vec{r} = \left[x^3 - \frac{x^4}{32} + \frac{17x^6}{128} \right]_0^2$$

Step 8: Evaluate the limits at $x = 2$ and $x = 0$

Substitute the upper limit (2) and subtract the lower limit (0):

$$\int_C \vec{F} \cdot d\vec{r} = \left((2)^3 - \frac{(2)^4}{32} + \frac{17(2)^6}{128} \right) - 0$$

Calculate the powers of 2:

$$(2)^3 = 8, \quad (2)^4 = 16, \quad (2)^6 = 64$$

Substitute these values back into the numeric expression:

$$\int_C \vec{F} \cdot d\vec{r} = 8 - \frac{16}{32} + \frac{17(64)}{128}$$

Simplify each fraction term:

$$\frac{16}{32} = \frac{1}{2}$$

$$\frac{64}{128} = \frac{1}{2} \implies \frac{17(64)}{128} = \frac{17}{2}$$

Combine the remaining fractions over their shared denominator:

$$\int_C \vec{F} \cdot d\vec{r} = 8 - \frac{1}{2} + \frac{17}{2}$$

$$\int_C \vec{F} \cdot d\vec{r} = 8 + \frac{16}{2}$$

$$\int_C \vec{F} \cdot d\vec{r} = 8 + 8 = 16$$